KPIT | APEX LAB

POSIX Pipes

Inter-Process Communication (IPC) using Anonymous Pipes

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What Is a Pipe?

A unidirectional communication channel between processes

- Data flows **like a queue**: first-in, first-out (FIFO)
- Created using **pipe() system call**
- Exists in memory (**kernel buffer**)
- Commonly used between **parent-child processes**

Anonymous Pipes Features

- Not associated with a file name
- Only work between related processes(parent-child or siblings created via fork())
- Automatically destroyed when both ends close
- Provide half-duplex communication (one-way)

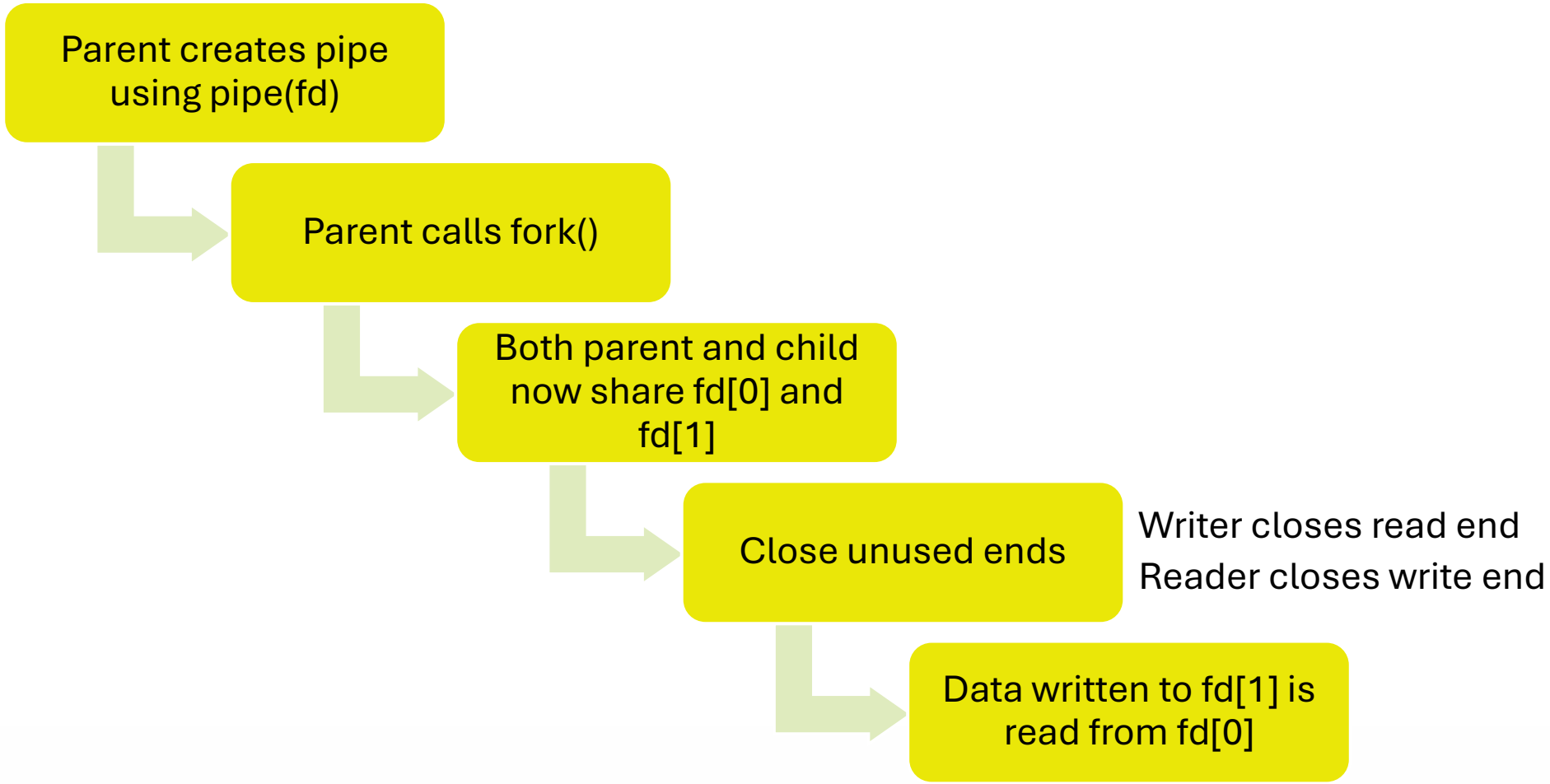
Anonymous pipes enable simple, fast IPC within a family of processes.

pipe() System Call

- int pipe(int fd[2]);
- Returns:
  - fd[0]: Read end
  - fd[1]: Write end
- On success: returns 0
- On failure: returns -1, sets errno

- Common errors:
  - EMFILE: Too many file descriptors
  - ENFILE: System FD limit reached

Pipe Lifecycle



Example 1: Parent Writes, Child Reads (1/2)

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <string.h>
#include <sys/wait.h>

int main(void)
{
int pipefd[2]; // pipefd[0] = read-end, pipefd[1] =
write-end
pid_t pid;
const char *msg = "Hello from parent!\n";

if (pipe(pipefd) == -1) {
        perror("pipe");
        exit(EXIT_FAILURE);
}
```

Example 1: (2/2)

```
pid = fork();
if (pid == 0) { // ---------- CHILD ----------
        close(pipefd[1]); // close unused write end

        char buffer[128];
        ssize_t nbytes = read(pipefd[0], buffer,
        sizeof(buffer));
        if (nbytes == -1) {
                perror("read");
                exit(EXIT_FAILURE);
        }
        buffer[nbytes] = '\0';
        // NUL-terminate for safe printing

        printf("Child received: %s", buffer);
        close(pipefd[0]); // done reading
        _exit(EXIT_SUCCESS);
}
```

```
else { // ---------- PARENT ----------
        close(pipefd[0]); // close unused read end

        ssize_t len = strlen(msg);
                if (write(pipefd[1], msg, len) != len) {
                perror("write");
                exit(EXIT_FAILURE);
        }
        close(pipefd[1]); // signal EOF to child

        int status;
        waitpid(pid, &status, 0); // optional: reap child
        if (WIFEXITED(status))
                printf("Child exited with status %d\n",
        WEXITSTATUS(status));
        else
                 printf("Child terminated abnormally");
}

return 0;
}
```

Example 2: Child Writes, Parent Reads (1/2)

```
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <string.h>
#include <sys/wait.h>

int main(void)
{
int pipefd[2]; // pipefd[0] = read end, pipefd[1] = write end
pid_t pid;
const char *msg = "Hello from child!\n";

/* Create the pipe */
if (pipe(pipefd) == -1) {
perror("pipe");
exit(EXIT_FAILURE);
}
```

Example 2: (2/2)

```
pid = fork();

if (pid == 0) { // ---------- CHILD ----------
close(pipefd[0]); // child does not read -> close
read end

size_t len = strlen(msg);
if (write(pipefd[1], msg, len) != (ssize_t)len) {
perror("write");
exit(EXIT_FAILURE);
}

close(pipefd[1]); // send EOF to parent
_exit(EXIT_SUCCESS);
}
```

```
else { // ---------- PARENT ----------
        close(pipefd[1]); // parent does not write -> close write end

        char buffer[128];
        ssize_t n = read(pipefd[0], buffer, sizeof(buffer));
        if (n == -1) {
                perror("read");
                exit(EXIT_FAILURE);
        }
        buffer[n] = '\0'; // NUL-terminate for safe printing

        printf("Parent received: %s", buffer);
        close(pipefd[0]); // done reading

        int status;
        waitpid(pid, &status, 0); // reap child
        if (WIFEXITED(status))
                printf("Child exited with status %d\n",
                WEXITSTATUS(status));
        else
                printf("Child terminated abnormally");
}
return 0;
}
```

Two-way Communication Using Two Pipes (1/3)

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <errno.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MSGSIZE 128

int main(void)
{
	int pipe_parent_to_child[2];
	// Parent writes, child reads
	int pipe_child_to_parent[2];
	// Child writes, parent reads
```

```c
/* Create both pipes */
if (pipe(pipe_parent_to_child) == -1 ||
pipe(pipe_child_to_parent) == -1) {
	perror("pipe");
	exit(EXIT_FAILURE);
}
```

Two-way Communication Using Two Pipes (2/3)

```
pid_t pid;
pid = fork();
if (pid == -1) {
perror("fork");
exit(EXIT_FAILURE);
}

/* ------------- CHILD PROCESS ------------- */
if (pid == 0) {
        /* Close unused ends */
        close(pipe_parent_to_child[1]);
        // close write end (parent→child)
        close(pipe_child_to_parent[0]);
        // close read end (child→parent)
```

```
/* Read a message from the parent */
char buf[MSGSIZE] = {0};
read(pipe_parent_to_child[0], buf, MSGSIZE - 1);
printf("Child received: \"%s\"\n", buf);

/* Prepare a reply */
const char *reply = "Hi parent, message received!";
write(pipe_child_to_parent[1], reply, strlen(reply) + 1);
/* Clean up */
close(pipe_parent_to_child[0]);
close(pipe_child_to_parent[1]);
_exit(EXIT_SUCCESS);
}
```

Two-way Communication Using Two Pipes (3/3)

```
/* ------------ PARENT PROCESS ------------ */
else {
/* Close unused ends */
close(pipe_parent_to_child[0]);
// close read end (parent→child)
close(pipe_child_to_parent[1]);
// close write end (child→parent)

/* Send a message to the child */
const char *msg = "Hello child, this is parent.";
write(pipe_parent_to_child[1], msg, strlen(msg) + 1);
/* Receive child's reply */
char reply_buf[MSGSIZE] = {0};
read(pipe_child_to_parent[0], reply_buf, MSGSIZE - 1);
printf("Parent received: \"%s\"\n", reply_buf);
```

```
/* Clean up */
close(pipe_parent_to_child[1]);
close(pipe_child_to_parent[0]);

/* Wait for child to exit */
wait(NULL);
}

return 0;
}
```

Blocking Behavior of Pipes

When read blocks:	When write blocks:	When read returns 0:	When write fails with SIGPIPE:
• Pipe empty • Write end still open	• Pipe buffer full	• All write ends closed → EOF	• All read ends closed → writer gets SIGPIPE

Pipe Buffering

- Pipes use kernel buffer (size ≈ 64 KB on modern Linux)
- Writes ≤ PIPE_BUF are atomic
- Good for logging or sending structured messages

Closing Pipe Ends Correctly

Golden Rules:

- Close **unused ends immediately** after fork
- Writers must close when done → allows readers to detect EOF
- Readers must close when done → prevents writers from blocking forever

Incorrect closure → deadlocks & hangs

Pipes in Embedded Systems

Typical use cases:

- Logging process receiving data via a pipe
- Sensor poller → filter thread (as separate process)
- Motor control command-forwarder
- Inter-process pipelines without using shared memory

Troubleshooting & Common Errors

- Deadlock due to not closing unused ends
- Reader blocked forever (writer didn't close)
- Writer SIGPIPE (reader closed early)
- Reading partial messages without framing
- Forgetting to flush stdout when writing from parent/child

Summary

- Pipes: simplest form of IPC
- Unidirectional & anonymous
- Use with fork() for parent-child communication
- Understand blocking, EOF, atomic writes
- Use two pipes for duplex communication

Scenario: Parent–Child Line Filter with Two-Way Control (1/4)

Context

- Build a line filter as a pair of related processes:
- The parent streams input lines to the child via a pipe;
- The child transforms lines (e.g., uppercase or ROT13) and sends results back to the parent through a second pipe.
- Parent prints transformed lines and supports a runtime mode switch command.

Scenario: Parent–Child Line Filter with Two-Way Control (2/4)

PART 1 — Plumbing & Direction

- Create two pipes:
  - p2c: parent → child (write at parent, read at child)
  - c2p: child → parent (write at child, read at parent)
- fork(); close unused ends properly in each process.

Scenario: Parent–Child Line Filter with Two-Way Control (3/4)

PART 2 — Transform Logic (Child)

- Default transform: uppercase.
- Read a line from p2c, transform, write to c2p.
- Exit on EOF when parent closes its write end.

Scenario: Parent–Child Line Filter with Two-Way Control (4/4)

PART 3 — Control Channel (Parent)

- Parent reads stdin:
  - Normal lines → send to child, read transformed reply, print.
  - Command :mode rot13 → send control token __CMD_MODE_ROT13__ to child; child switches mode.
  - Command :quit → close p2c write end; read remaining output; exit.